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OPERATING SYSTEM (PROJECT REVIEW – 1)

TOPIC :- Secure Capability-Based File System

1. Problem Statement

Traditional file systems mainly use users, groups, and permission bits for resource protection. In a multi-process environment, this may not provide sufficiently fine-grained control over which process can perform a particular operation on a particular resource.

The proposed project develops a **mini operating-system file system using capability-based access control**, where each process is explicitly assigned capabilities specifying the resources it can access and the operations it can perform. Every access request is verified before execution, preventing unauthorized operations.

2. Objective

- Develop a **virtual file system** with basic file and directory operations.
- Create and manage simulated processes.
- Implement **capability-based access control** for individual processes.
- Support READ, WRITE, DELETE, and EXECUTE permissions.
- Allow dynamic **granting and revocation** of capabilities.
- Maintain an **audit log** of allowed and denied access attempts.
- Demonstrate OS concepts of **processes, file systems, protection, and security**.

3. Proposed Approach / Solution

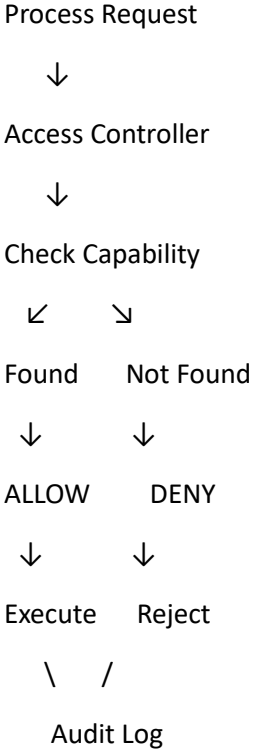
The project will implement a mini OS-like file system simulator with capability-based access control. The system will maintain simulated processes, a virtual file system, and a capability table that defines which operations each process is allowed to perform on specific files. Whenever a process requests an operation such as read, write, delete, or execute, the Access Controller will verify the corresponding capability. The operation will be permitted only if the required capability exists; otherwise, it will be denied. The system will also support granting and revoking capabilities dynamically and maintain an audit log of all successful and failed access attempts.

Module	Function
Process Manager	Creates processes and assigns unique PIDs.
Virtual File System	Manages files, directories, and file operations.
Capability Manager	Creates, stores, checks, grants, and revokes capabilities.
Access Controller	Verifies capabilities before every resource operation.
Audit Logger	Records successful and unauthorized access attempts.

A capability will be represented as:

Process + Resource + Allowed Operation

When a process requests an operation:



5. Technologies

Language: C++
Interface: Command-Line Interface
Compiler: GCC/G++
Development: VS Code / any C++ IDE
Data Structures: Classes, structures, vectors/maps

6. OS Concepts Covered

Processes • File Systems • Protection • Access Control • Resource Security • System Calls • Least Privilege

7. Expected Outcome

The final system will simulate a secure OS file system in which processes can access resources **only through explicitly assigned capabilities**. The system will successfully distinguish between authorized and unauthorized operations, support dynamic permission management, and maintain an audit trail of resource-access requests.

Scope: The project is a simulation of relevant OS components and does not attempt to implement a complete kernel or hardware-level operating system.